For instance, the binary number is $A_3\ A_2\ A_1\ A_0 = 0010$, its corresponding one's complement number is $\overline{A}_3\ \overline{A}_2\ \overline{A}_1\ \overline{A}_0 = 1101$

The same principle will apply for number having any bit length and its corresponding one's complement number can be obtained by complement each bit.

8.5.2 Two's Complement Method

The two's complement of a binary number is the number that results when we add '1' to the one's complement number. The formula for two's complement of a binary number is given below.

Two's complement number = one's complement +1

For instance, to find the two's complement number of 0101, the following procedure is employed.

$0101 \Rightarrow 1010$ (1's *complement*)

$1010+1 \Rightarrow 1011$ (2's *complement*)

8.5.3 Binary Addition

For binary addition, the arithmetic rules used are given below.

1. 0+0=0
2. 0+1=1, No carry is formed
3. 1+0=1, i.e. carry=0
4. 1+1=0,with a carry of 1, and sum = 0 i.e. $1+1 = 10_2$.

This is a binary number10 and not the decimal number ten. Here, the first digit 0 is called sum and next digit 1 is called carry.

Examples:

1. Add the binary numbers 1011 and 1100

$$\begin{array}{r} 1011+ \\ 1100 \\ \hline 10111 \end{array}$$

Sum=0111, and Carry=1

2. Add the binary number 11101 with 11011001.

The first number = 00011101+
The second number = 11011001
Result = 11110110

8.5.4 Binary Subtraction

The general rules for carrying out the binary subtraction are given below.

1. 0-0=0
2. 1-0=1
3. 1-1=0
4. 0-1=1,with a borrow of 1 from the next higher bit.
5. 10-1=1

Example: Subtract 0111 from 1011

$$\begin{array}{l} 1011 \Rightarrow 11_{10} \\ 0111 \Rightarrow 7_{10} \\ \hline 0100 \Rightarrow 4_{10} \end{array}$$

First column => 1-1 =0

Second column =>1-1 = 0

Third column =>0-1 = 10-1 = 1

Forth column => After borrowing, the fourth column becomes 0

Hence, 0-0=0

$1011_2 - 0111_2 = 0100_2$

8.6 BINARY CODES

All digital circuits operate with only two states namely, High and Low or ON and OFF or 1 and 0. In binary number system, the number of bits required goes on increasing as the numbers become larger and larger. So, some special binary codes are required to represent alphabets and special characters. Based on these points, different types of binary code have been developed.

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